Introduction to Dynamic Programming

Workshop – Term 1, Week 8

Susan He, Austin Song, Matthew Gu

Outline

1 Housekeeping

2 We have to go deeper

3 On being not a goldfish

4 More dimensions???

5 Appendix and resources

Attendance!

Please scan the attendance form so you can get free food

VJudge For Workshop

If you're confident with the content for today, feel free to get started on problems!

Link: https://vjudge.net/contest/802029

Password: `cpmsoc`

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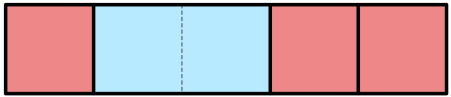

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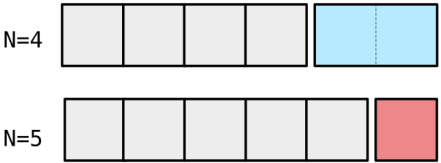

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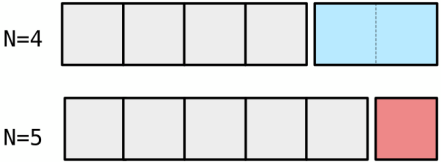

Thus, we have the recurrence relation $A(n) = A(n-1) + A(n-2)$ for $n > 2$.

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Base cases: $A(0)$ and $A(1)$. From this, we can generate all other $A(n)$.

Terminology

- Recursion: When a function calls/depends on itself.
- Base case: The simplest subproblem of the function. In our case, $N = 0, 1$.
- State: the variables that define a specific subproblem. This is the n in $A(n)$.
- State transitions: How states relate to each other, $A(n) = A(n-1) + A(n-2)$ for $n > 2$.

Implementation

Demo time!

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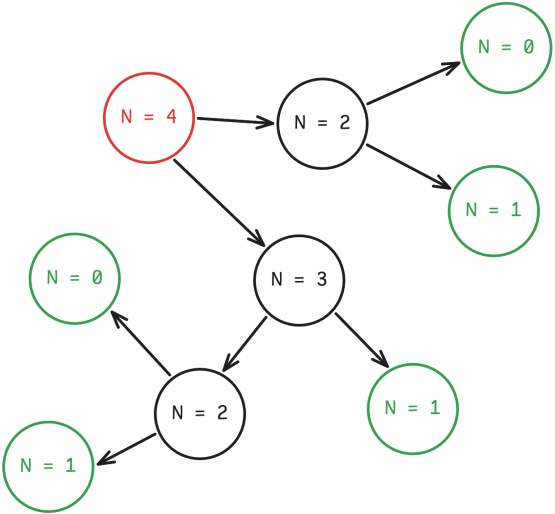

What's the obvious problem?

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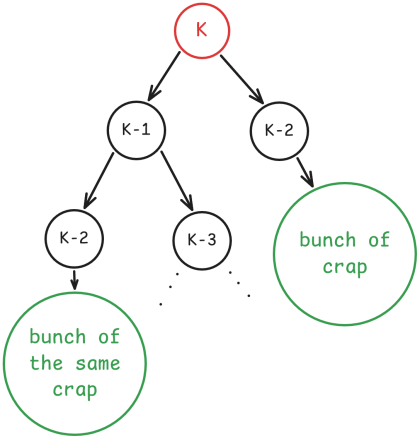

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Let's find out by testing it – why not!

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Recursion combined with memory is known as **dynamic programming** (DP).

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P.S. If you've done COMP2521, you'll recognise that recursive DP is basically DFS! Then, you don't need to think bottom-up to realise why the time complexity improves so much.

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If these things are true, we can just chuck a `dp[state]` array over our recursion and steal the free optimisation.

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Constraints:

- $N \leq 10^5$, $K \leq 10^2$.

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Now that we have a recursive formulation, we just chuck DP over it and win :)

Exercise: What's the final time complexity?

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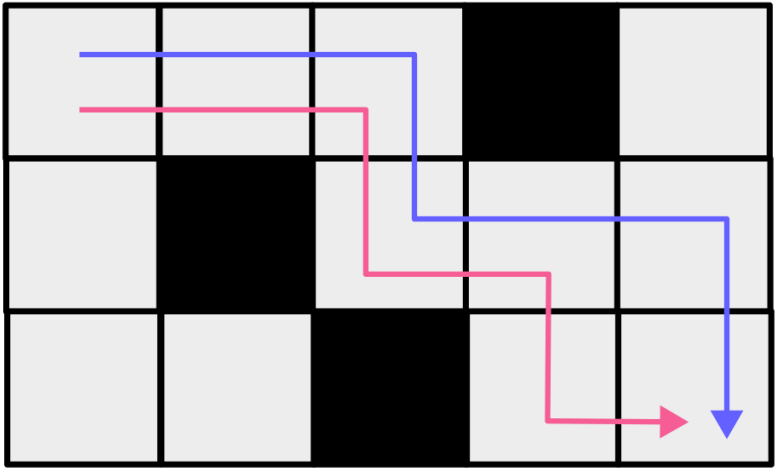
Unfortunately, D. Piddy is under investigation by the federal police. If $p_{i,j} = 1$, it means police are present at cell (i, j), and will arrest anyone they see going to the party. How many ways are there to reach Piddy's mansion, while avoiding police detection?

Visualisation



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$$dp(i,j) = \begin{cases} 0 & \text{if } p_{ij} = 1 \\ dp(i-1,j) + dp(i,j-1) & \text{if } p_{i,j} = 0 \end{cases}$$

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What do we need in our state?

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Exercise: What's the final time complexity?

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Samples

Sample Cases

Sample Input:

```
8
1 3 4 7 9 10 11 14
```

Sample Output:

```
4
```

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- Our time complexity is $O(N^2)$.

Outline

1 Housekeeping

2 We have to go deeper

3 On being not a goldfish

4 More dimensions???

5 **Appendix and resources**

Feedback!

Please fill out feedback form so we know how to run future workshops :D

(a) Feedback form

(b) Attendance form

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